

Women's World Cup Statistical Analysis

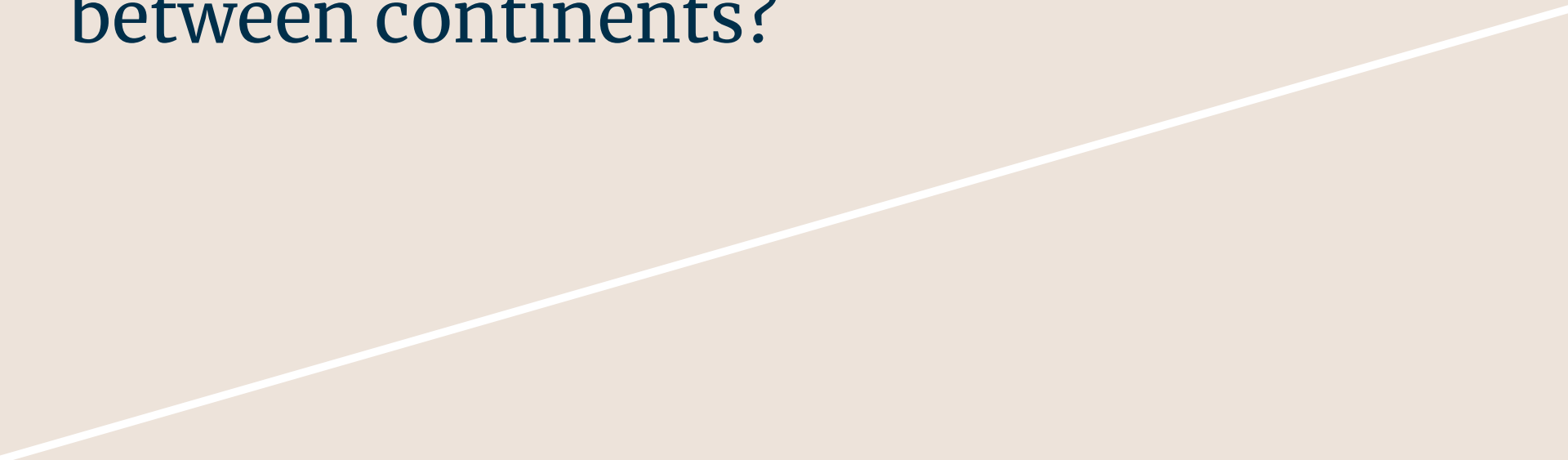
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Dataset Introduction

- Women's World Cup statistics from 1991–2019
- Population data
- 136 observations of 22 variables, including year, squad, age, matches played, goals per game, and others
- Created three new categorical variables: win, continent, and stage

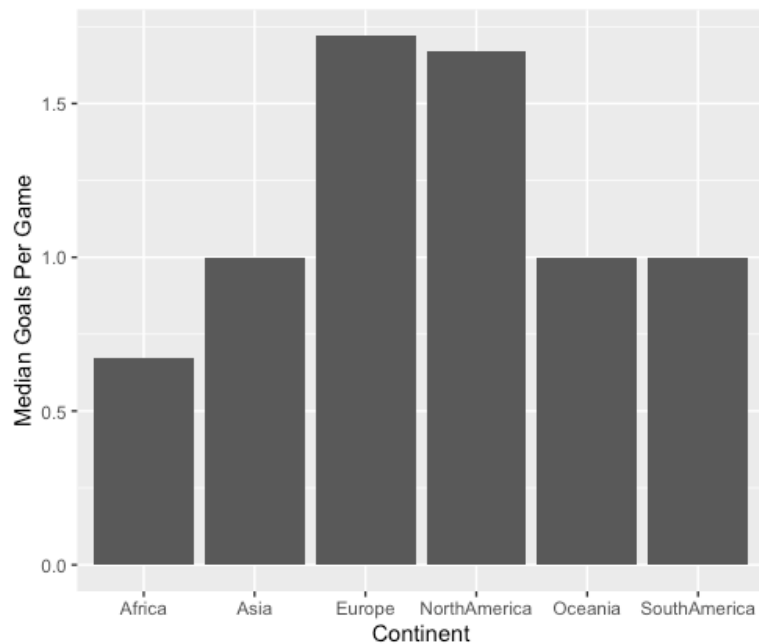
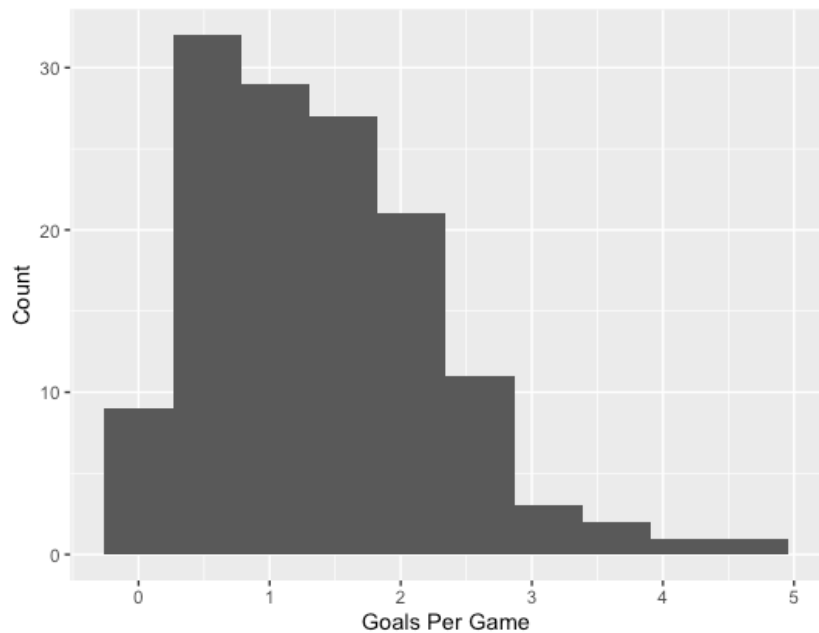
Question 1: Is there a difference in the median number of goals scored per game between continents?



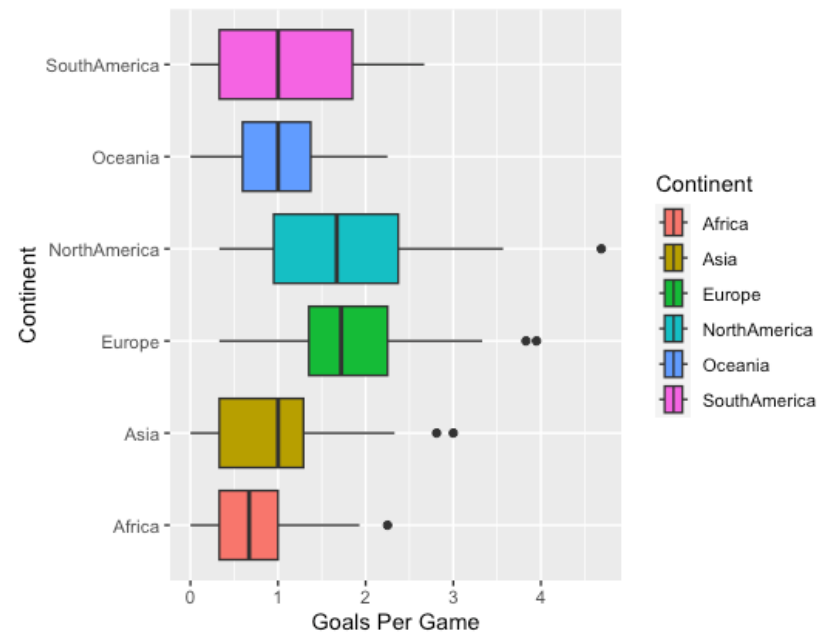
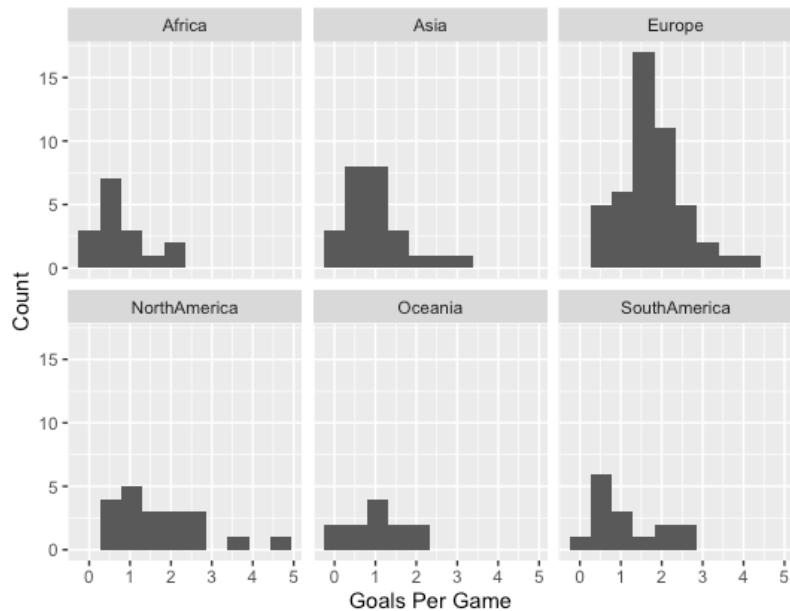
Introduction

- Data was broken down into levels using the “continent” variable
- Conditions for ANOVA test were not met, so we’re using the Kruskal-Wallis test
- This test compares medians between groups instead of means because the data is skewed
- Determines if there is a statistically significant difference in median number of goals scored per game between continents

Exploratory Data Analysis



Exploratory Data Analysis



Conditions

- We can assume independence if we treat the Women's World Cup data as a sample of women's soccer as a whole
- The data appears to be heavily right skewed, so it doesn't meet the normality condition
- Constant variance is satisfied among and between groups
- Since not all conditions are met for the ANOVA test, we have to use the Kruskal-Wallis instead

Hypotheses and Results

- Hypotheses:
 H_0 : The medians are the same among all continents
 H_A : At least one median is different
- The H statistic is 28.267 and the p-value is 0.00003
- Reject the null hypothesis and conclude that the median is different for at least one continent

Post-Hoc Comparison

- Wilcoxon test compares individual pairs of medians to determine which groups are different
- We have a new significance level that we get from the Bonferroni correction, which in this case is 0.003
- P-value for Europe and Africa: 0.00087
- P-value for Europe and Asia: 0.00189
- These continents have a statistically significant difference in the median number of goals scored per game

Question 2: Which variables are the best predictors of winning the Women's World Cup?



Introduction

- Winning the World Cup is a binomial response variable
- The “win” variable was coded to account for this
- Must use logistic regression
- The model predicts the probability of a win or loss based on given values for the explanatory variable

Logistic Regression Model

- Equation for the model is
 $\text{logit}(p_i) = -30.7328 + 3.1510 * (\text{goals per game}) + 0.7979 * (\text{age})$
- AIC = 30.26
- Higher number of goals scored per game and higher average player age leads to a higher probability of winning the World Cup

Predictions

World Cup Year	Winner	Probability of Winning
1991	United States	0.9167
1995	Norway	0.7240
1999	United States	0.3953
2003	Germany	0.9629
2007	Germany	0.7120
2011	Japan	0.0082
2015	United States	0.1456
2019	United States	0.9706

- We know the actual results of each of the past World Cups, so we can check the accuracy of our model
- Very accurate for 5 past winners
- Model is most accurate for teams that scored more goals per game and have a higher average player age

Inference

- Statistical significance of predictors in the model can help to determine the accuracy of our model

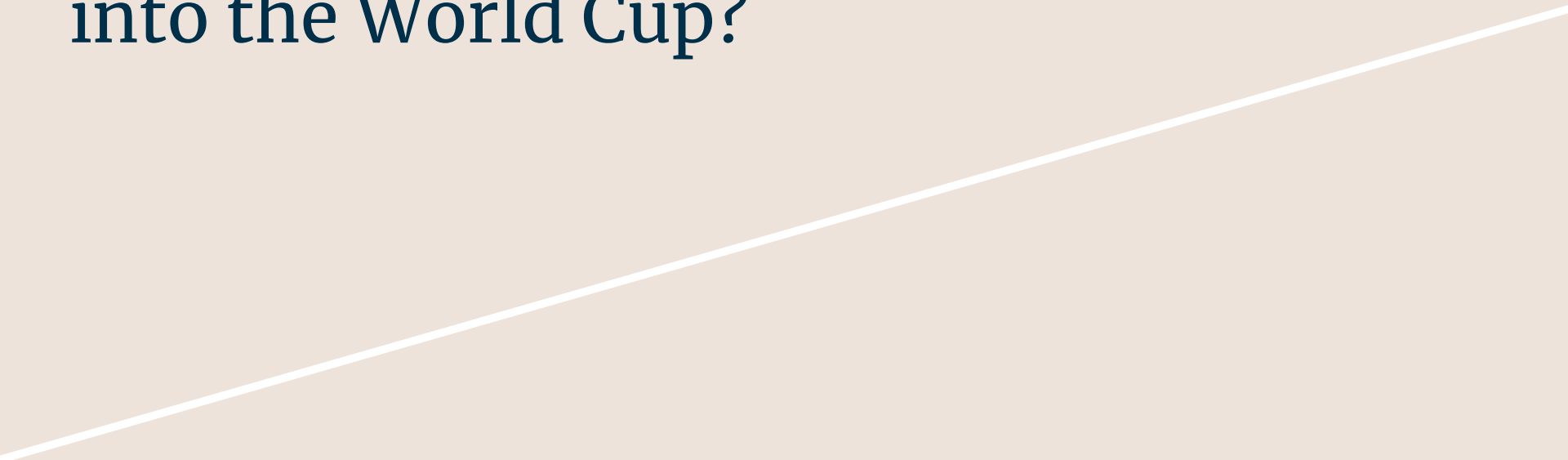
- Hypotheses:

H_0 : Slope coefficient for goals per game is zero, given that age is still included in the model

H_0 : Slope coefficient for age is zero, given that goals per game is still included in the model

- P-value for goals per game: 0.00052
- P-value for age: 0.0679

Question 3: Which variables are the best predictors of how far a team makes it into the World Cup?



Introduction

- The stage of the World Cup a team reaches is an ordinal categorical variable
- Teams were assigned to levels based on how many games they played in the tournament
- Coded using the “stage” variable
- Ordinal logistic regression is formatted the same way as logistic regression, except the response variable has more than two levels

Ordinal Logistic Regression Model

- Predictor variables included goals per game and age
- $AIC = 180.0013$
- There are two intercepts, one for going from the Group Stage to the Knockout and the other from Knockout to Final
- Slope coefficient for goals per game is 2.1162 and slope coefficient for age is 0.3095

Predictions

Country	Group Stage	Knockout	Final
United States	0.0016	0.0881	0.9103
France	0.0746	0.7560	0.1694
Jamaica	0.9053	0.0930	0.0017

- Predictions based on data from the 2019 World Cup
- Model gives a probability for each team making it to a certain stage of the tournament
- Gives validity to the model

Inference

- Similar to logistic regression, the statistical significance of predictor variables helps us to determine the accuracy of the model
- Hypotheses:

H_0 : Slope coefficient for goals per game is zero, given that age is still included in the model

H_0 : Slope coefficient for age is zero, given that goals per game is still included in the model

- P-value for goals per game: ~ 0
- P-value for age: 0.0053